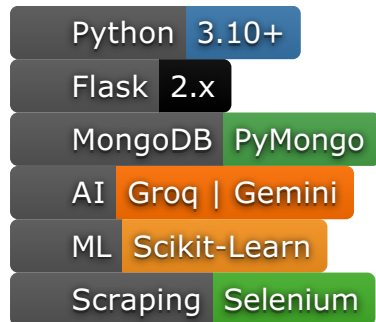


AI-Powered CS Course Recommender & Intelligence Dashboard

An intelligent, full-stack web application that serves as a comprehensive career and educational companion for Computer Science students and professionals. By merging advanced Large Language Models (LLMs), dynamic data visualization, real-time scraping, and interactive mock interview simulations, the platform acts as a personalized mentor—guiding users from learning new skills to acing technical and behavioral interviews.



Exhaustive Feature Breakdown

1. AI-Powered Course Recommender & Study Planner

The core of the educational platform, designed to eliminate the guesswork from upskilling.

- **Skill-Based Recommender Engine:** Users input their current skills and career goals. The AI parses the database to recommend the exact courses required to bridge the gap.
- **Dynamic AI Study Plans:** Generates a custom, week-by-week curriculum and roadmap to help users organize their learning journey based on their availability and targets.
- **User Authentication & Profiles:** Secure session-based login system that tracks user progress, saves their generated study plans, and associates analytics data directly to their profile via MongoDB.

2. The Interview Analyzer (Mock Interview Platform)

A fully integrated, multi-modal interview simulator designed to test both hard algorithmic skills and soft behavioral skills using state-of-the-art LLMs.

- **Generative Search Mode:** Users can search for any CS topic (e.g., "Docker Networking"), and the AI instantly generates relevant theoretical questions and their full explanations on the fly, bypassing static databases entirely.
- **Technical Flashcards with Ace Editor:** Replaces standard text boxes with a professional, syntax-highlighted IDE (Ace Editor). Supports Python, JavaScript, Java, and C++.
- **Algorithmic AI Evaluation:** When code is submitted, the backend LLM (Groq Llama-3-70b) acts as a strict technical interviewer. It explicitly grades the code on **Time Complexity (Big-O)**, **Space Complexity**, and identifies **edge cases** the user may have missed.
- **Behavioral Mode (Voice & STAR Method):** Integrates WebRTC to allow users to verbally record their answers to behavioral questions (e.g., "Tell me about a time you failed").
- **Whisper API Transcription:** Audio is captured as .webm and sent to Groq's **Whisper-Large-V3**

API for blazing-fast transcription.

- **STAR Scorecard:** The LLM strictly evaluates transcribed answers using the **STAR Method** (Situation, Task, Action, Result), returning a 1-10 score for each individual component and an overall critique.
- **Dynamic Follow-up Questioning:** If an answer is correct but lacks depth, the AI generates context-aware follow-up questions to simulate the pressure of a real human interviewer probing for details.
- **Historical Progress Radar:** All interview scores (Technical, Behavioral, System Design, etc.) are securely logged to MongoDB. The dashboard uses **Chart.js** to render an interactive Radar chart, allowing users to track their proficiency growth over time.

3. Semantic Search Engine & AI Rating System

A powerful fallback engine that indexes scraped course data using traditional machine learning and NLP.

- **TF-IDF Vectorization & Cosine Similarity:** Employs TF-IDF with bigram support for context-aware search, ranking courses based on true semantic relevance rather than just keyword matching.
- **AI-Powered Star Ratings:** Uses **NLTK VADER Sentiment Analysis** to score course descriptions and student reviews.
- **Review-Weighted Scoring:** When student reviews are available, ratings are heavily weighted (70%) based on the sentiment of real student feedback.
- **Platform Prestige Bonus:** Courses from elite institutions (MIT, Google, Stanford, IBM) receive an artificial prestige boost in the algorithm to guarantee high-quality results.
- **NLP Review Summarization:** Extracts up to 10 reviews per course from Coursera, tokenizes them, and performs frequency analysis to generate human-readable summaries (e.g., *"Based on 8 reviews, students frequently mention: algorithms, difficult, rewarding"*).

4. Intelligence EDA (Exploratory Data Analysis) Dashboard

A dedicated visualization center for data-driven career decisions.

- **Market & Skill Visualization:** Features dynamic, interactive charts (powered by Chart.js/Plotly) that help users understand trending tech stacks, high-demand skills, and salary analytics.
- **Course Distribution Analytics:** Visualizes the spread of course difficulties, providers, and ratings across the scraped database.

5. Admin Dashboard & Multi-Platform Web Scraper

The backbone of the platform's data ingestion pipeline.

- **Secure Admin Portal:** A protected route allowing administrators to manage the underlying dataset.
- **CSV Data Uploads:** Admins can manually upload standard CSV files containing course data, which the backend safely parses and inserts into MongoDB.
- **Deep Web Scraping Engine:** Uses headless Chrome (Selenium) to scrape 24+ targets across 5 platforms (MIT OCW, Coursera, Cybrary, Khan Academy, CourseTalk).
- **Asynchronous Execution:** Scraping jobs run on background threads, preventing server lockups, while a real-time progress bar updates the UI dynamically.
- **robots.txt Compliant:** Respects crawling policies to ensure ethical data gathering.

Modern UI & UX Design

- **Glassmorphism Aesthetic:** Features translucent panels, soft glows, and blurred backdrops for a

premium, native-app feel.

- **Fully Responsive & Accessible:** Fluid layouts adapt seamlessly to mobile, tablet, and desktop screens.
- **Dynamic Theming:** Smooth transitions between high-contrast Light Mode and deep Dark Mode.
- **Generative UI:** Micro-animations, loaders, and **Lucide Icons** respond instantly to AI streaming states.

How the AI Engine Works

Structured JSON Enforcement & Prompt Engineering

To ensure the AI acts as a reliable software component rather than an unpredictable chatbot, all backend LLM prompts strictly enforce JSON formatting. By injecting rigid schemas into the system prompts (e.g., forcing keys for "time_complexity", "star_score", and "followup"), the Flask backend can safely parse and bind the AI's intelligence directly to UI components without relying on fragile regex hacks.

Tech Stack

Layer	Technology
Backend Framework	Python Flask, Threading (Async Processing)
Frontend Languages	HTML5, CSS3, Vanilla JavaScript
Frontend Libraries	Ace Editor (IDE), Chart.js (Radar/EDA), Lucide (Icons)
Artificial Intelligence	Groq API (Llama-3-70b-8192, Whisper-Large-V3), Google Gemini 2.5 Flash
Natural Language Processing	NLTK (VADER Sentiment, Tokenization, Stopwords)
Machine Learning	Scikit-Learn (TF-IDF + Cosine Similarity)
Web Scraping	Selenium (Headless Chrome) + BeautifulSoup4
Data Storage	MongoDB (PyMongo), JSON, Pandas

License

This project is open-source and intended for educational purposes.

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